



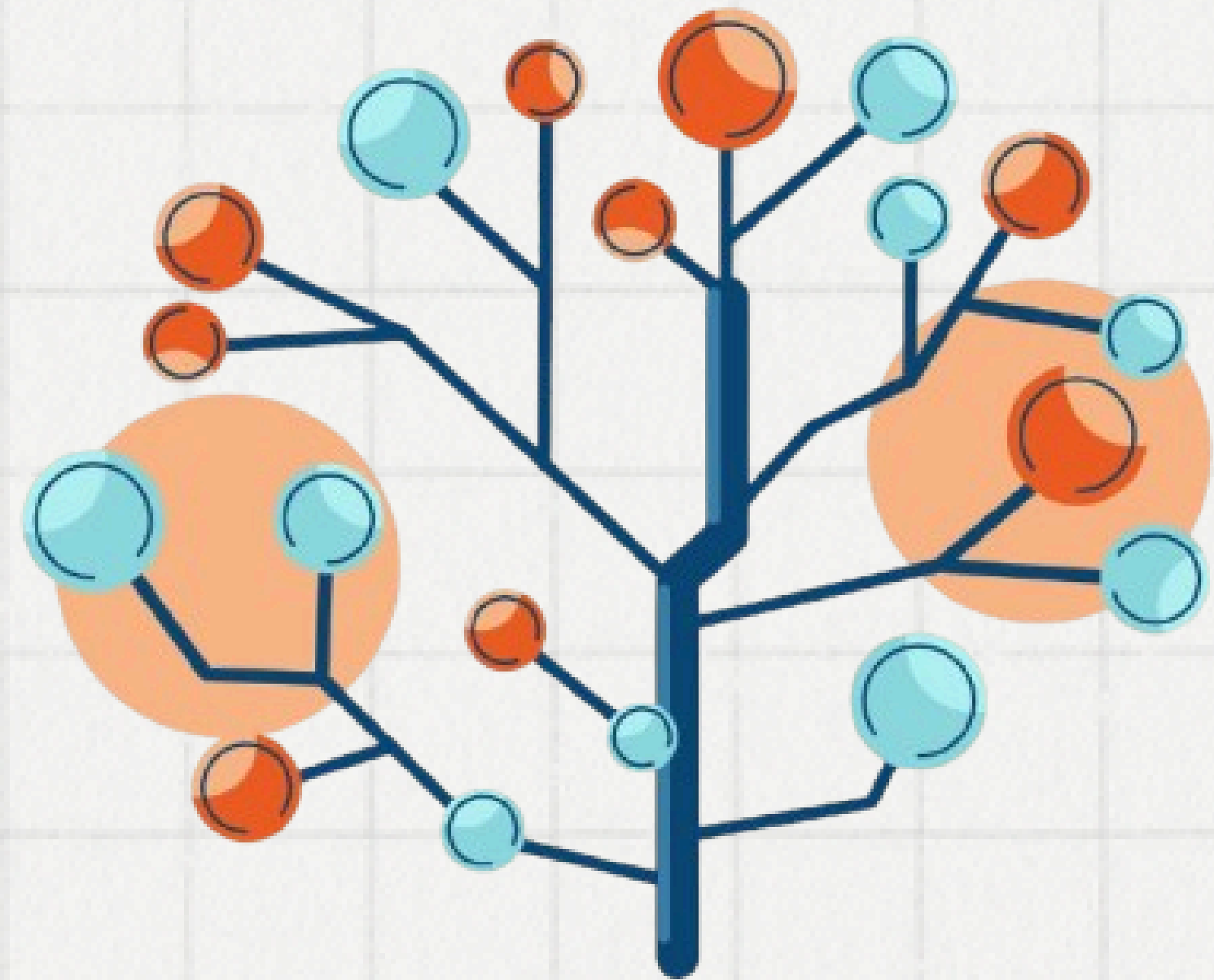
Decision Tree

1820222041 温富勝
1820222029 罗健豪
1820222030 林哲豪

Introduction

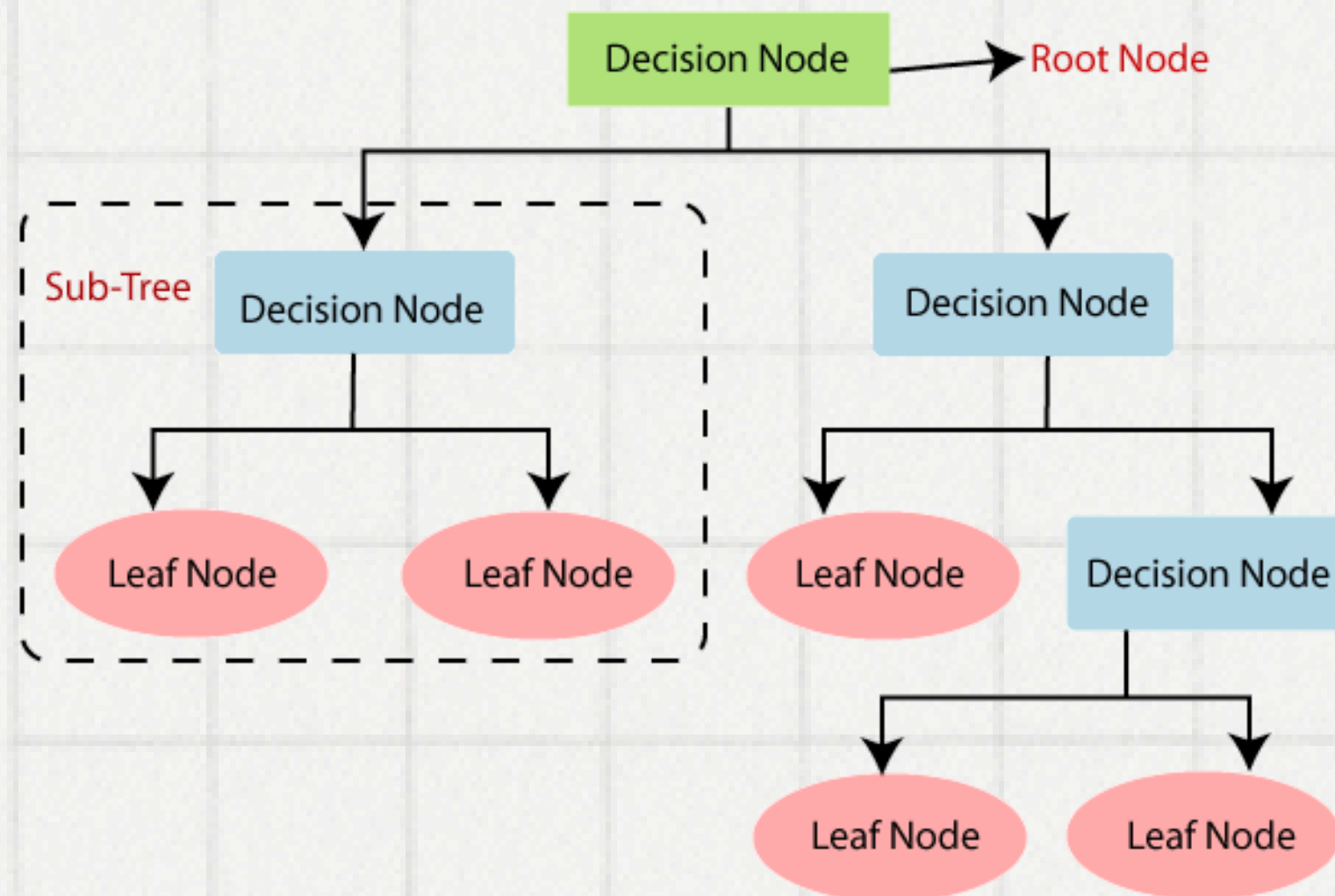
decision tree is supervised learning technique

- can be used for both classification and regression problems (mostly for classification)
- tree structured classifier (internal nodes – features of dataset, branches – decision rules, leaf node – outcome)
- graphical representation for getting all the possible solutions
- Decision tree is a graph to represent choices and their results in form of a tree.
- The nodes in the graph represent an event or choice and the edges of the graph represent the decision rules or conditions.
- It is mostly used in Machine Learning and Data Mining applications using Python.



DECISION TREE

Terminologies



Root Node

- where the decision tree starts,
- represent entire dataset
- get divided into two or more homogeneous sets

leaf node

- final output node
- tree cannot be segregated further

Splitting

- process of dividing decision node (root node) into subnodes
- according to the given conditions

Branch/subtree

- tree formed by splitting the tree

Pruning

- process of removing unwanted branches

parent/child node

- root node is parent node
- other nodes are called child nodes

Pre-Process

Gather Data

Admissions_to_DMHAS_Addiction_Treatment_by_Town_Year_and_Mon

Search in Sheet

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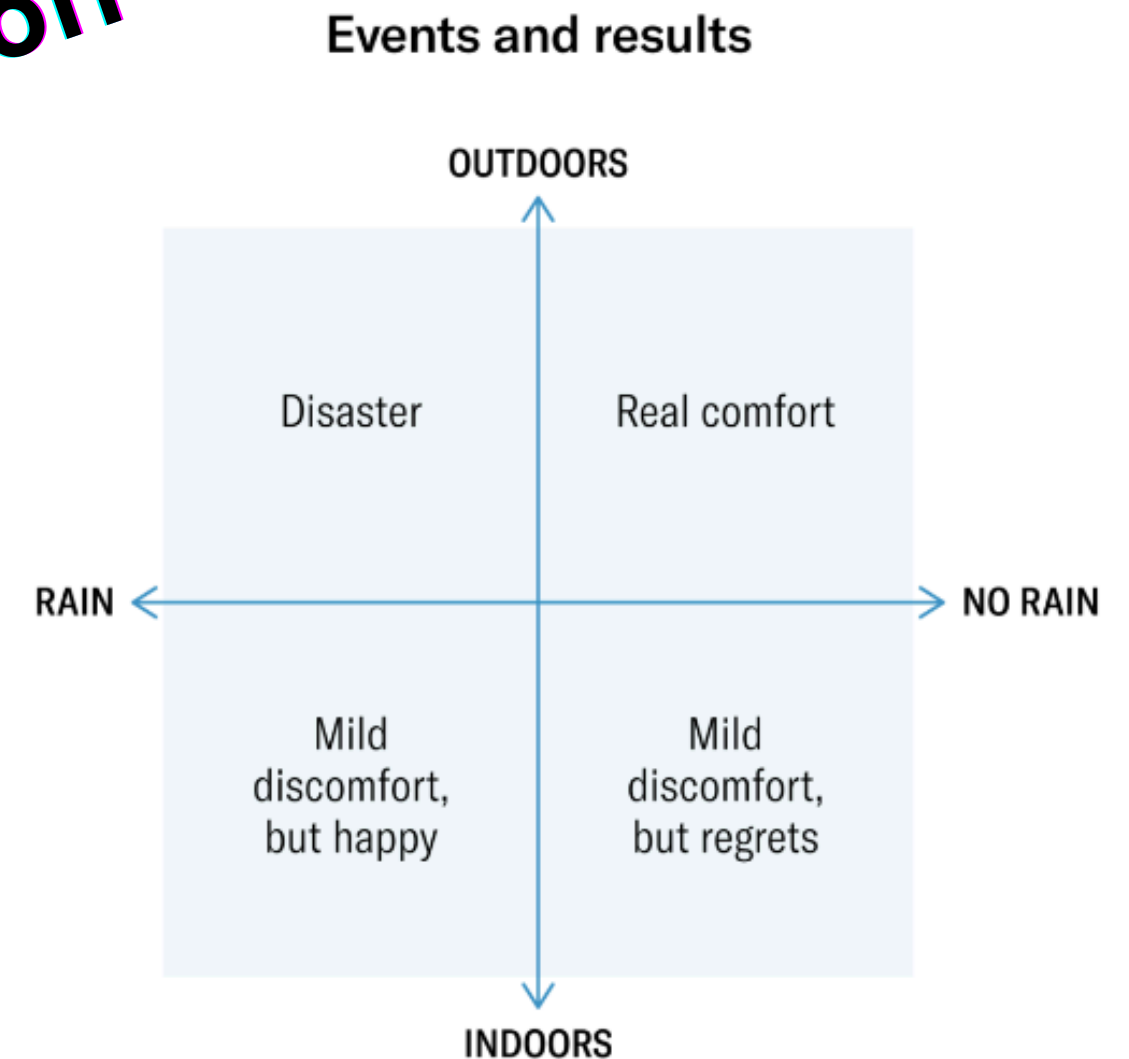
Edit Font Alignment Number Format

Paste Calibri (Body) 12 General Conditional Formatting Styles

A1 fx FiscalYear

	A	B	C	D	E	F	G
1	FiscalYear	MonthYear	Town	AdmMonth	FYMonthOrd	AdmYear	MonthTotal
2	2014	13-Jul	Ansonia	7	1	2013	42
3	2014	13-Aug	Ansonia	8	2	2013	43
4	2014	13-Sep	Ansonia	9	3	2013	39
5	2014	13-Oct	Ansonia	10	4	2013	33
6	2014	13-Nov	Ansonia	11	5	2013	38
7	2014	13-Dec	Ansonia	12	6	2013	45
8	2014	14-Jan	Ansonia	1	7	2014	35
9	2014	14-Feb	Ansonia	2	8	2014	38

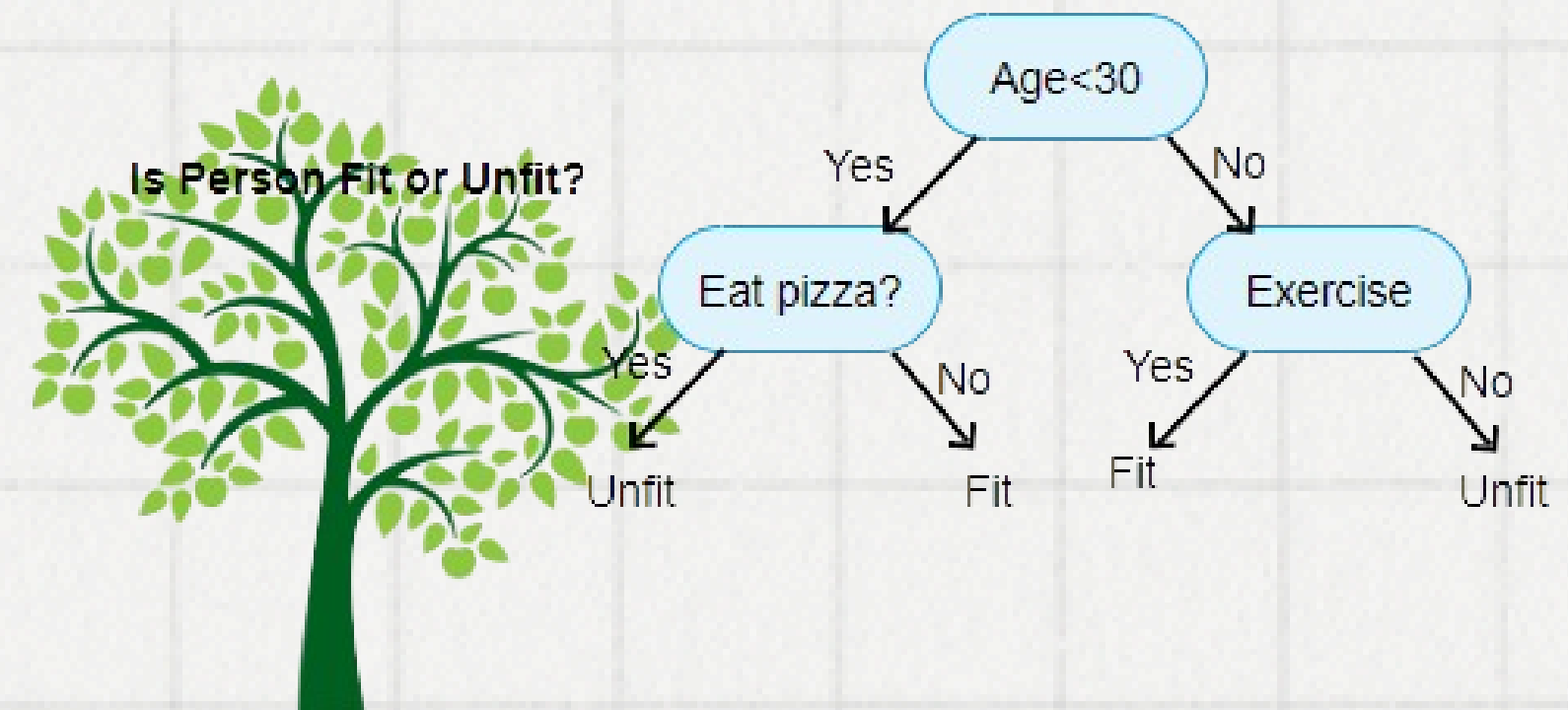
Payoff Table



How to select the best attributes?

Attribute Selection Measures

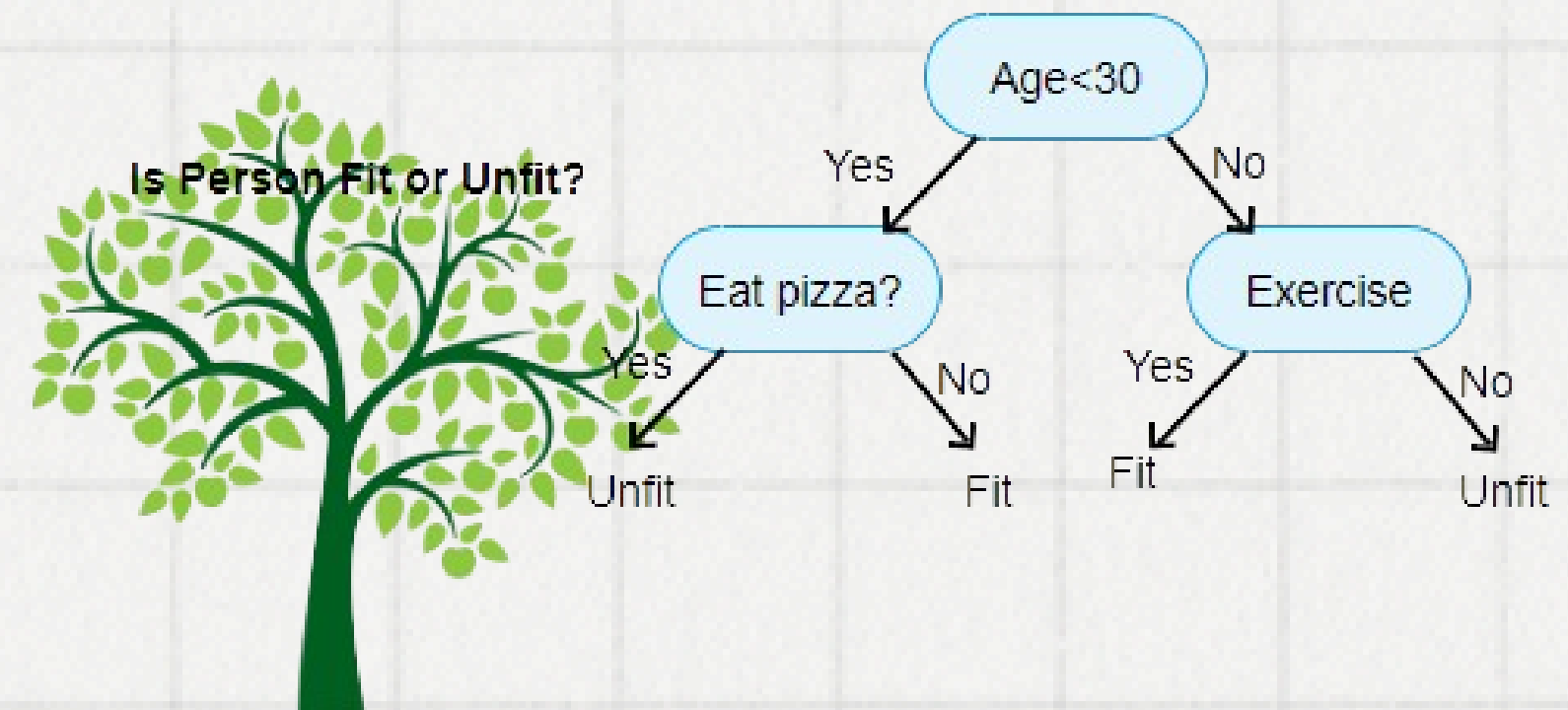
- Main issue that arise must be how to select the best attributes for root node and for sub-nodes.
- two techniques that can be used for ASM
 - Information Gain
 - measurement of changes in entropy after segmentation (how much information features a class)
 - entropy in this case is the randomness of each data
 - always try to maximize information gain, the highest information gain is split first



How to select the best attributes?

Gini Index

- Measure of impurity or purity used while making the decision tree
- low Gini index should be preferred compared to the high gini index
- a pure dataset is considered ideal as it can be easily divided into subsets of the same class
- it only creates binary splits



Algorithm

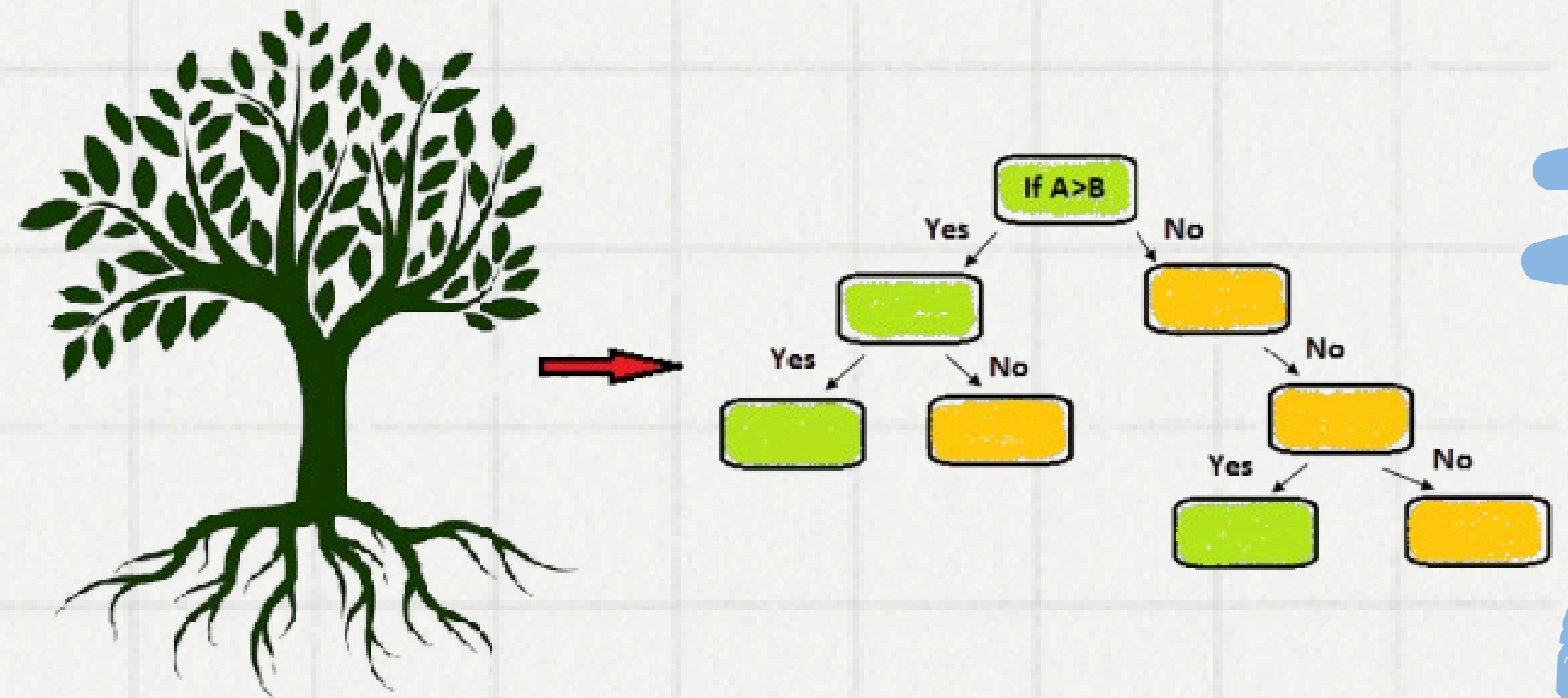
Step-1: Begin the tree with the root node, says S, which contains the complete dataset

Step-2: Find the best attribute in the dataset using Attribute Selection Measure (ASM).

Step-3: Divide the S into subsets that contains possible values for the best attributes.

Step-4: Generate the decision tree node, which contains the best attribute.

Step-5: Recursively make new decision trees using the subsets of the dataset created in step -3. Continue this process until a stage is reached where you cannot further classify the nodes and called the final node as a leaf node.



Strengths and Weakness

Strengths

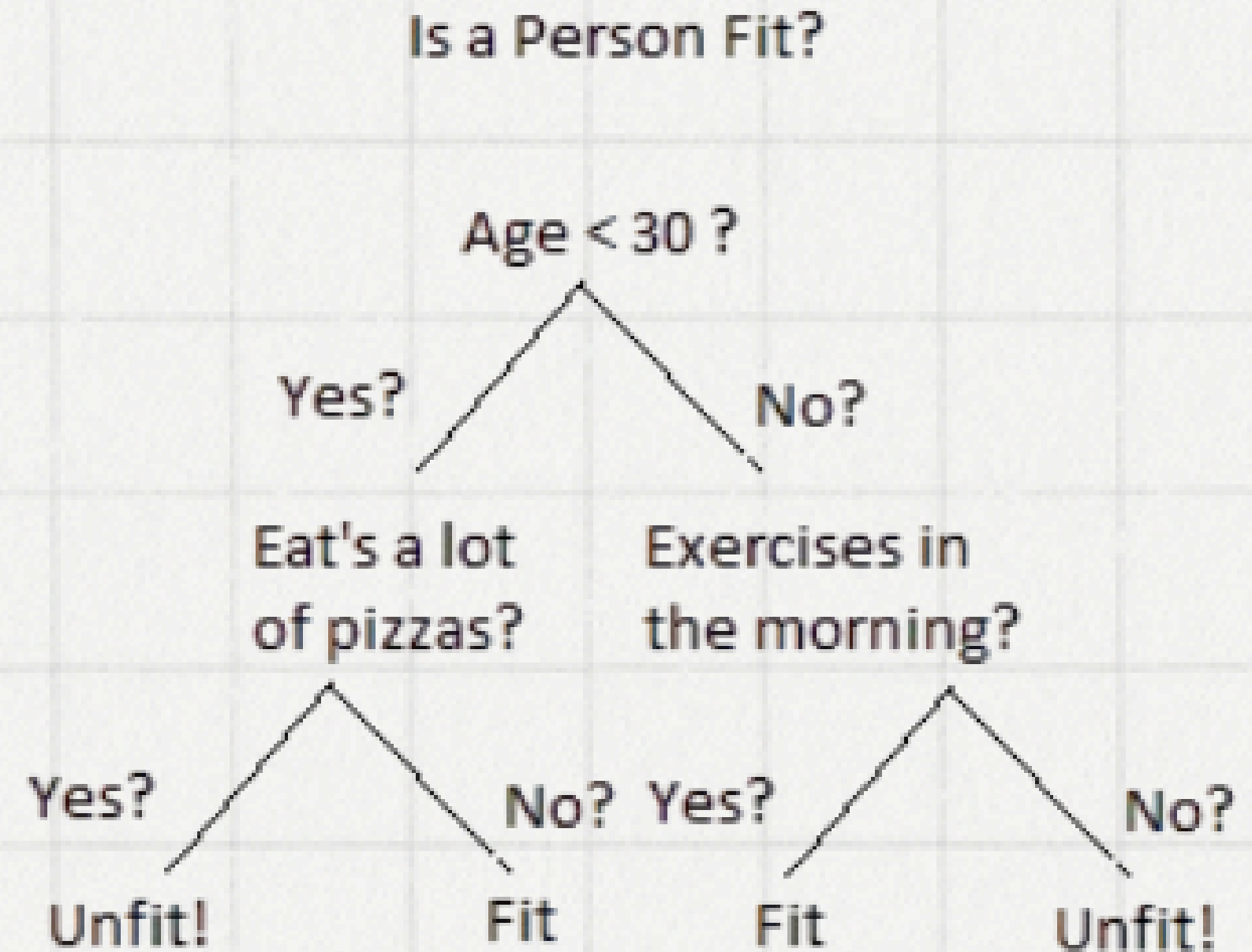
1. Simplicity
2. Versatility
3. Non-parametric
4. Feature Selection

Weakness

5. Overfitting
6. Instability
7. Bias
8. Limited with complexity

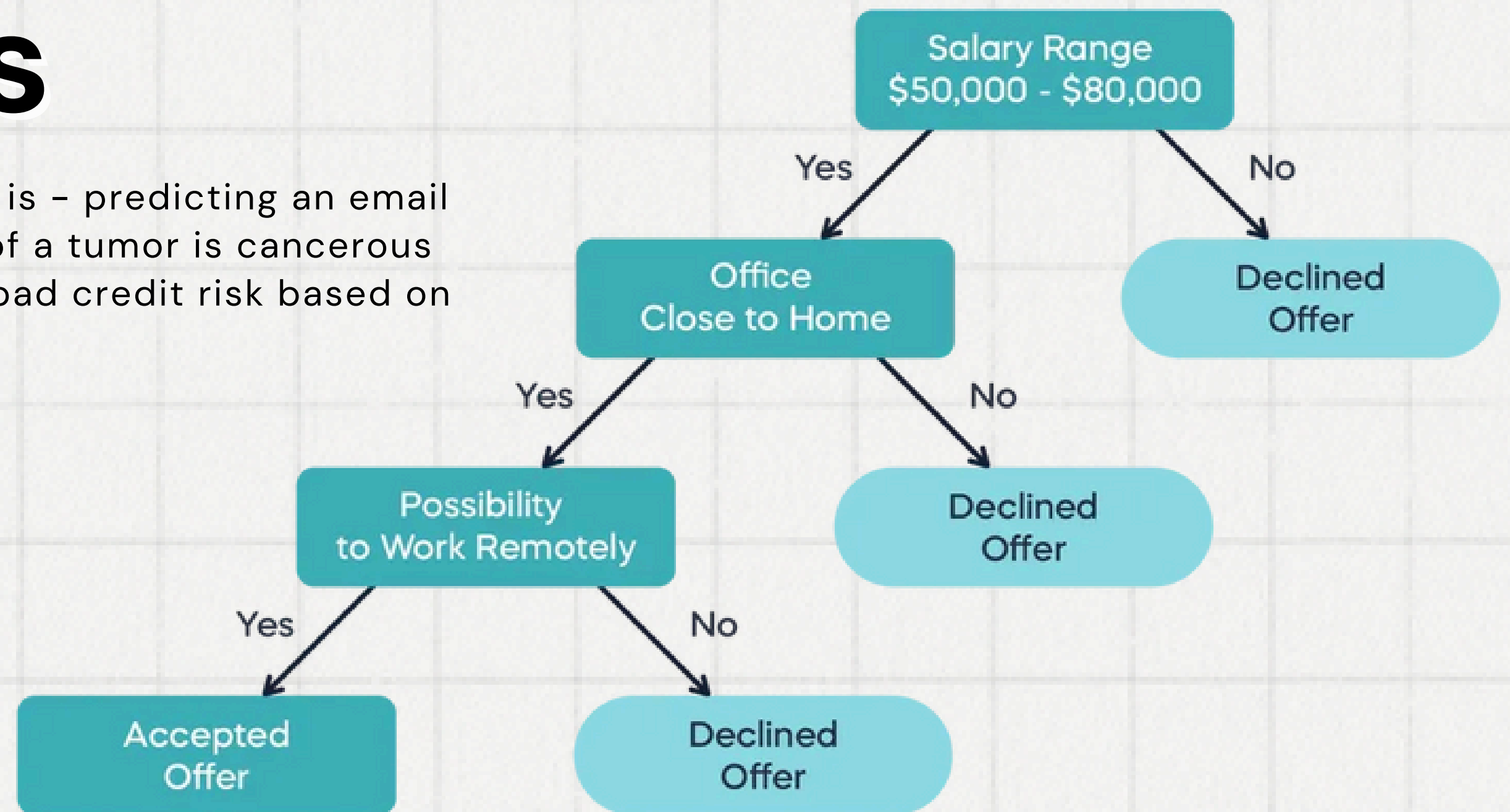
Limitation

- Trees can be very non-robust (unstable)
- not fit for continuous variable (variables that keeps on changing)
- greedy algorithms cannot guarantee the result for the most optimal decision tree
- can create biased trees if some classes dominate
- low prediction accuracy for a dataset
- very limited to one variable but in actually a lot of variables are affecting the results.



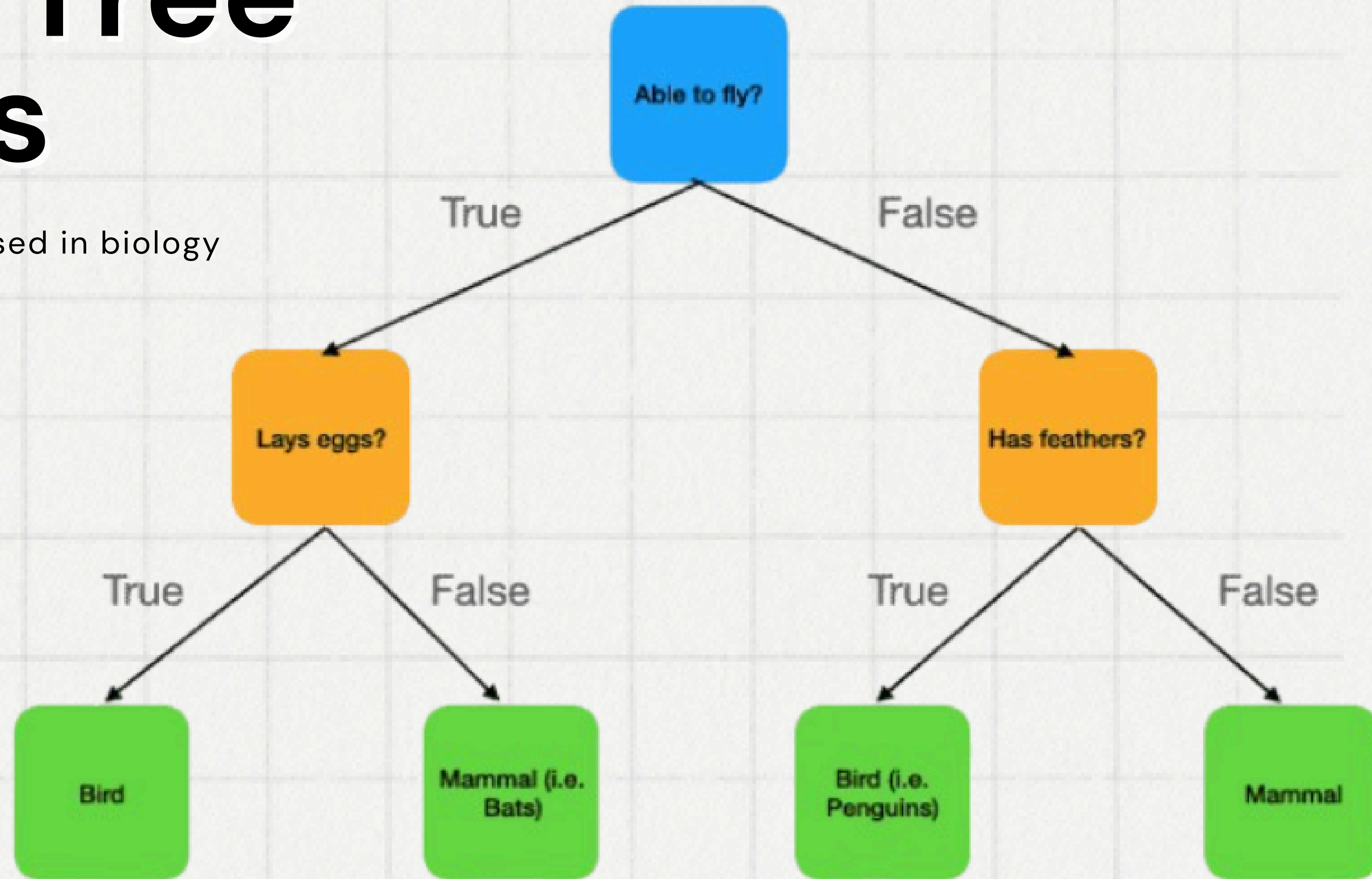
Decision Tree Examples

Examples of use of decision tree is – predicting an email as spam or not spam, predicting of a tumor is cancerous or predicting a loan as a good or bad credit risk based on the factors in each of these.



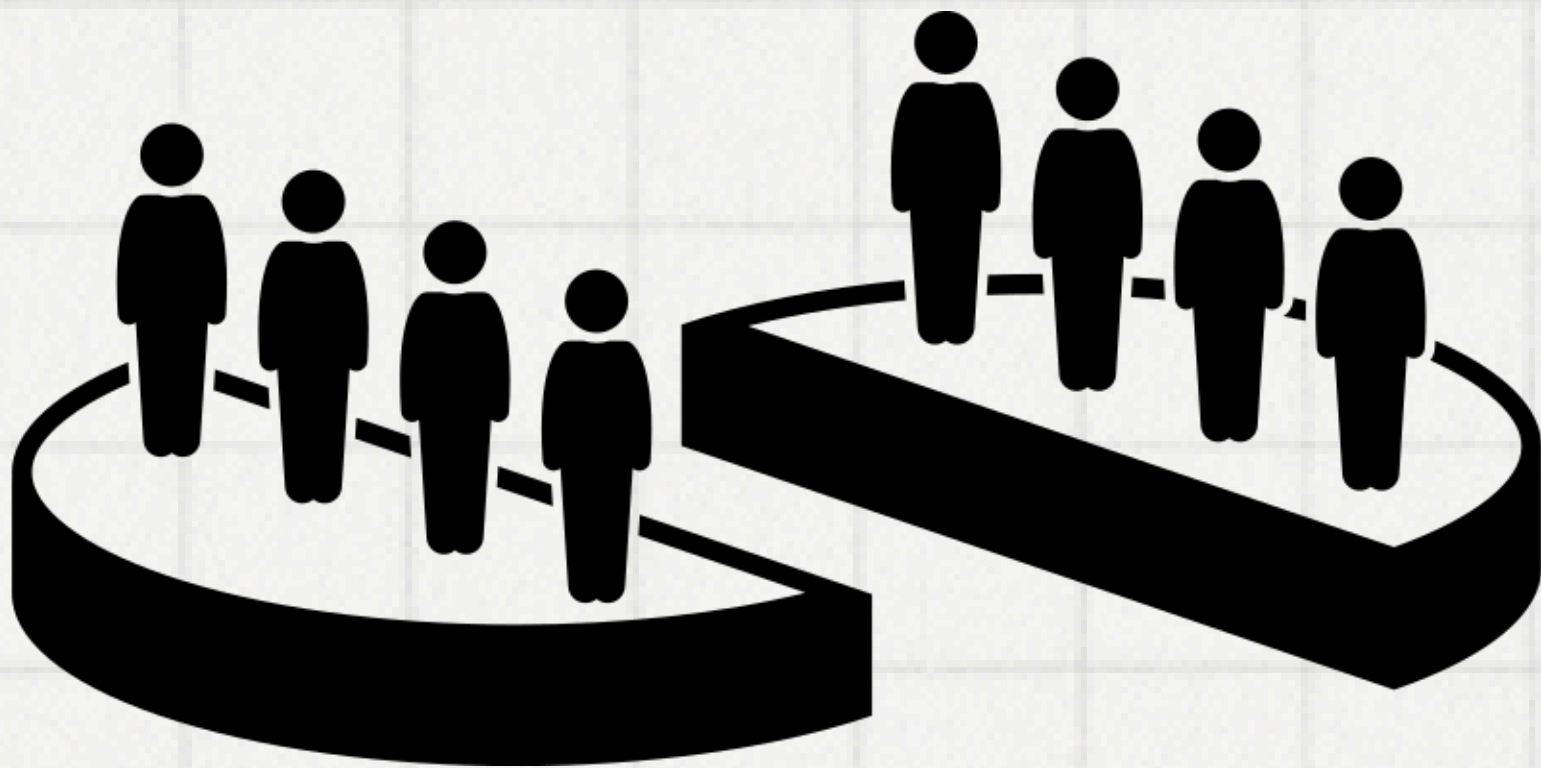
Decision Tree Examples

classification of animal (mainly used in biology to create the hierarchy)



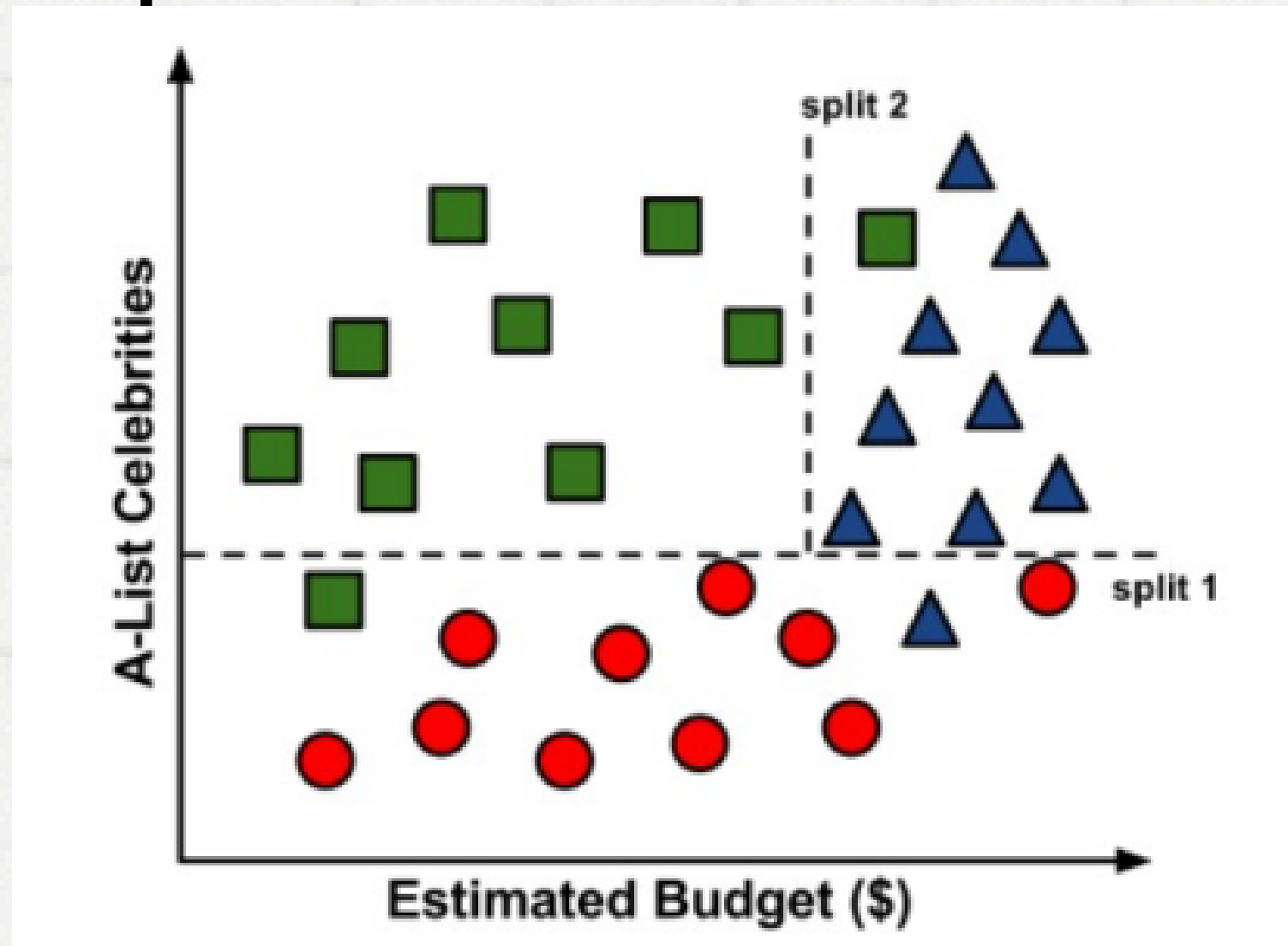
Divide and Conquer

Decision trees are built using a heuristic called recursive partitioning. • This approach is generally known as divide and conquer because it uses the feature values to split the data into smaller and smaller subsets of similar classes. • Beginning at the root node, which represents the entire dataset, the algorithm chooses a feature that is the most predictive of the target class. • The examples are then partitioned into groups of distinct values of this feature; this decision forms the first set of tree branches.

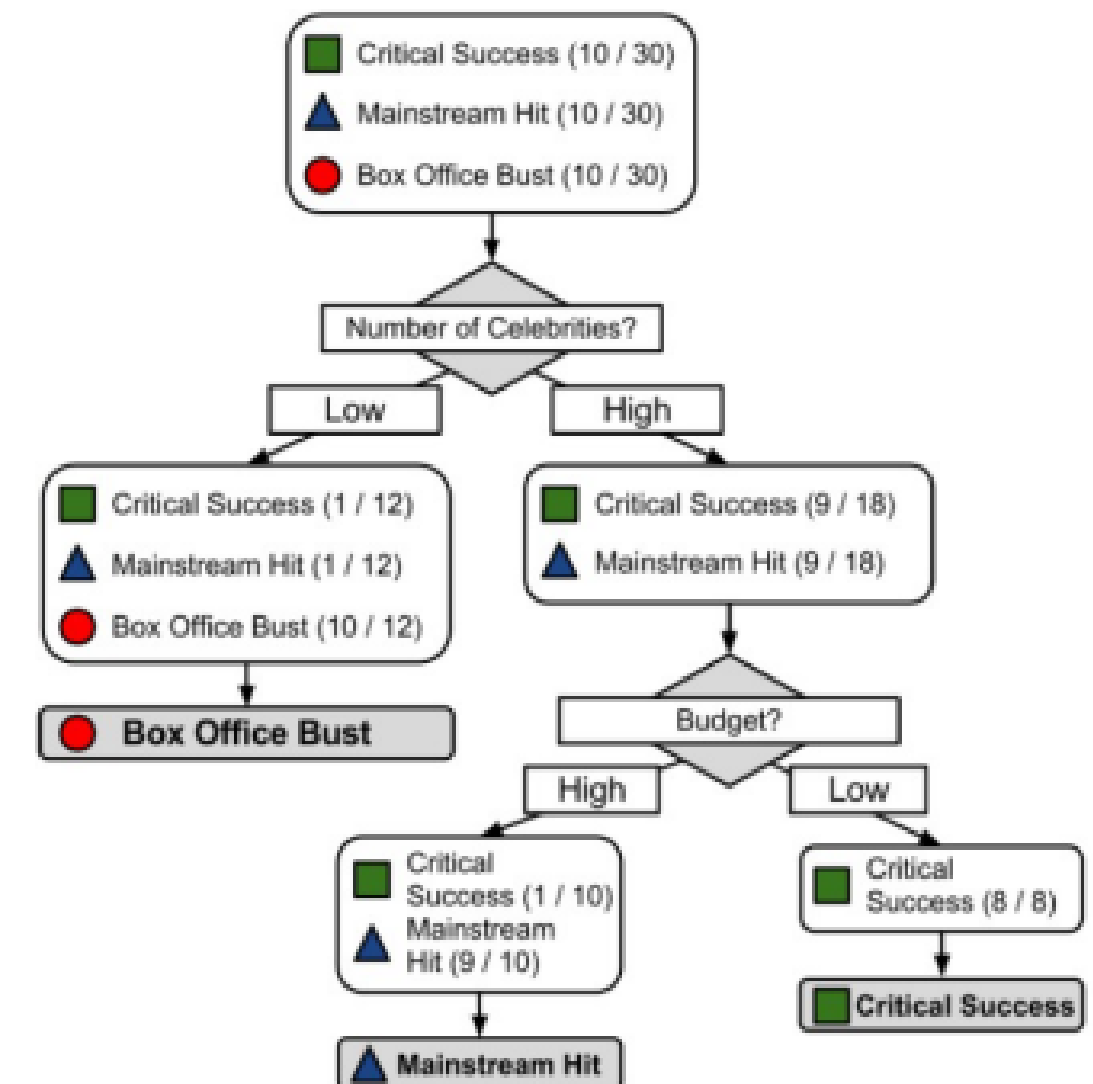


Example Case using Scatterplot

Step 3



Step 4



Packages needed

- **Data Analytics**
 - sudo pip3 install pandas
- **Decision Tree Algorithm**
 - sudo pip3 install sklearn
- **Visualization**
 - sudo pip3 install ipython
 - sudo pip3 install graphviz
 - sudo pip3 install pydotplus
 - sudo apt install graphviz



Reading Dataset

```
import pandas as pd
import numpy as np

dataset = pd.read_csv("banknotes.csv")
print dataset.shape    # Dataset's dimension
print dataset.head()   # First five entries

# Predictor variables
X = dataset.drop('Class', axis=1)

# Response variable
y = dataset['Class']
```

Training the Classifier

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.20) # Train: 80%, Test: 20 %

# Import the DT Classifier
from sklearn.tree import DecisionTreeClassifier

# Create Objects
classifier = DecisionTreeClassifier()

# Train the algorithm
classifier.fit(X_train, y_train)

# Predict test values
y_pred = classifier.predict(X_test)
```

Visualizing the Tree

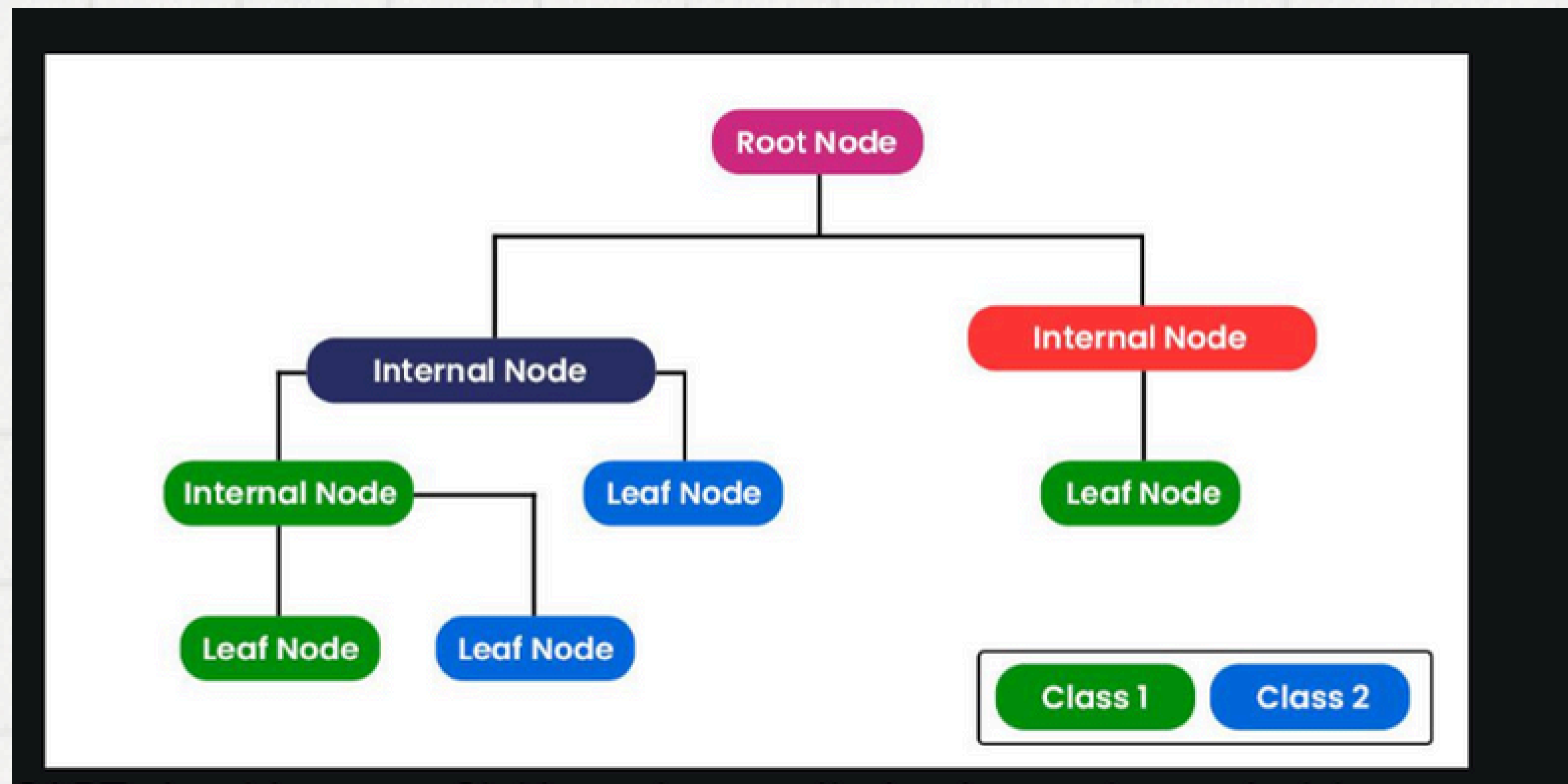
```
from sklearn.externals.six import StringIO
from IPython.display import Image
from sklearn.tree import export_graphviz
import pydotplus
dot_data = StringIO()
```

```
export_graphviz(classifier, out_file=dot_data,
filled=True, rounded=True, special_characters=True)
```

```
graph = pydotplus.graph_from_dot_data(dot_data.getvalue())
graph.write_png('tree.png')
```



CART (Classification and Regression Tree)



The background is a light blue grid with various hand-drawn blue doodles. These include several overlapping circles and loops at the top, a wavy line at the bottom center, and several checkmarks at the bottom right. There are also some abstract scribbles and lines scattered around the edges.

**Thank you
for
Listening**